CLIMATE DATA ANALYSIS · 1750–2024

Global CO_2 Emissions

Trends, Regional Shifts, and Per Capita Inequalities

A data-driven overview of how global carbon dioxide emissions have evolved from the Industrial Revolution to 2024, highlighting regional shifts, per capita disparities, and the latest annual changes.

AUDIENCE
Policy Analysts & Climate Negotiators

DATA SOURCE
Global Carbon Budget 2025 · NASA Mauna Loa

SCOPE
1750 – 2024

MAUNA LOA OBSERVATORY · FEB 2026

429

ppm CO_2

Atmospheric CO_2 Reaches 429 ppm

50% Above Pre-Industrial Levels. Ground-based and satellite measurements confirm the sharp acceleration, linked primarily to the burning of coal, oil, and natural gas.

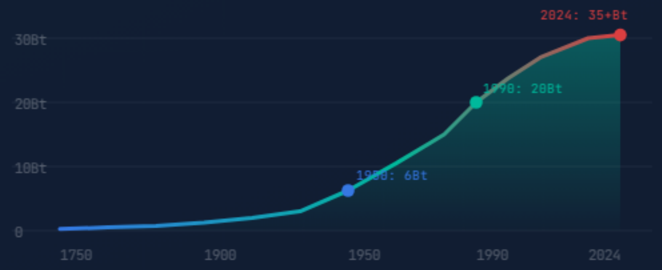
CONCENTRATION COMPARISON

Pre-industrial ~1750	~280 ppm
Today (Feb 2026)	429 ppm

Global Fossil Fuel CO_2 Emissions Grew Six-Fold

6 Billion to 35+ Billion Tonnes (1950–2024)

ANNUAL CO_2 EMISSIONS — GLOBAL TRAJECTORY



Source: Global Carbon Budget 2025 / Our World in Data

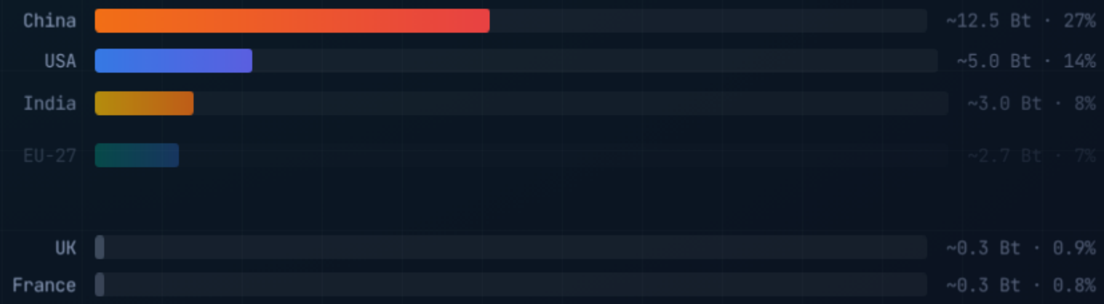
1950 — **-6 Bt**
Annual fossil fuel CO_2 — pre-acceleration baseline

1990 — **>20 Bt**
Nearly 4× the 1950 level — Cold War–era industrialisation

2024 — **35+ Bt**
Six-fold since 1950. Growth slowing — but not yet peaked

China Now Emits Over 12 Billion Tonnes

More than double the United States — annual totals, leading emitters



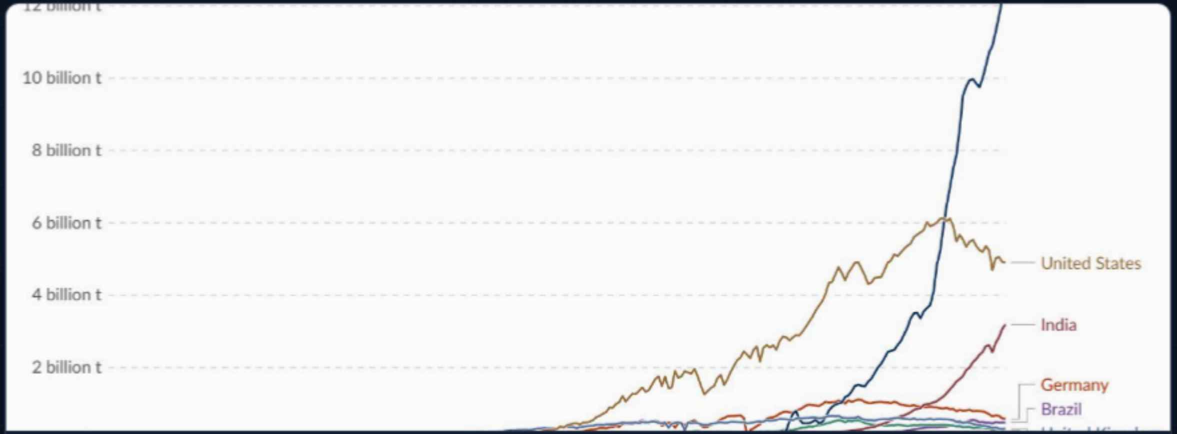


Fig. 3 — Annual CO_2 emissions by country

Asia Now Accounts for ~50% of Global Emissions

US + Europe Below One-Third — a dramatic geographic rebalancing

1900

Europe + US dominated. Total global emissions were small.

Europe + US	>90%
Asia	~7%
Other	~3%

By 1950: Europe + US still >85%

→

120 years of
industrialisation
in Asia & Global South

2024

Asia (~60% of world population) now produces half of all emissions.

Asia	~50%
US + Europe	<33%
Africa / LatAm / Other	~17%

Africa & South America each: ~3–4% (≈ intl. aviation & shipping)

Per Capita Gap: 15 t in the US vs. 0.1 t in Chad

A 150× Disparity — tonnes CO_2 per person per year

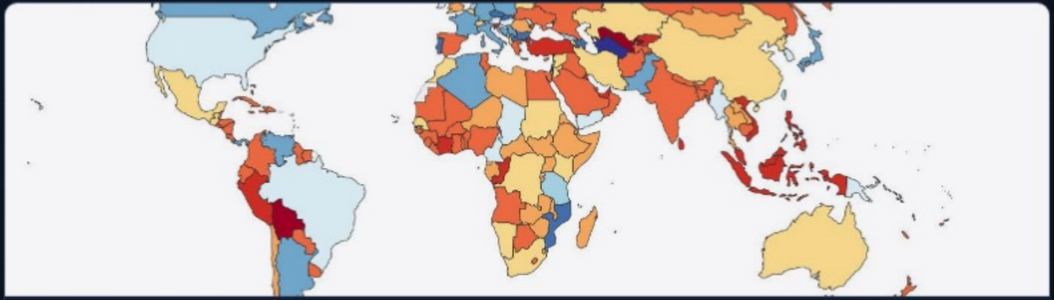


Fig. 1 — Per capita CO_2 over time

150× gap: The average American emits in under two days what the average person in Chad or Niger emits in an entire year.

2024 Snapshot: US and Europe Cut Emissions While Russia and SE Asia Increase

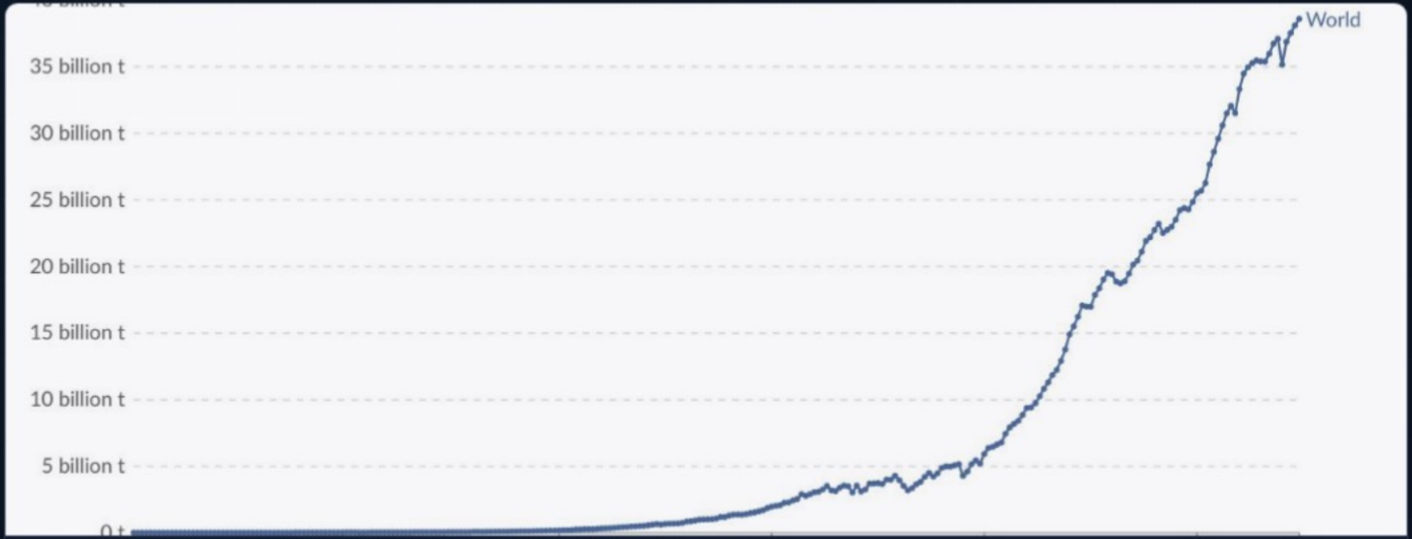


Fig. 4 — Annual % change in CO_2 emissions, 2024

Declining (Blue)

US and much of Europe: −1% to −5%. Continued clean energy transition, efficiency gains, and reduced industrial output in some sectors.

Rising (Orange/Red)

Russia, parts of Southeast Asia, and several African nations: +2% to +10% or more. Economic growth and fossil fuel expansion driving increases.

Large Jumps (>10%)

Some countries in South America recorded very large percentage increases (>10%), while others saw declines — a mixed regional picture.

Energy Mix Explains Why France Emits Far Less Per Capita Than Germany

COUNTRY	PER CAPITA CO_2	NUCLEAR SHARE	KEY ENERGY PROFILE
🇫🇷 France	~4 t	~70–75%	High nuclear + growing renewables → very low carbon grid
🇬🇧 United Kingdom	~4 t	~15%	Rapid wind & solar growth; gas declining; coal nearly phased out
🇩🇪 Germany	~8 t*	0% (2023)	Nuclear phase-out; heavy coal + gas reliance despite renewable growth
🇳🇱 Netherlands	~9 t*	<5%	Natural gas-heavy economy; major European gas hub

*EU-27 average ~7 t. Data: Global Carbon Budget 2025

Nuclear & Renewables Decarbonise the Grid

France's nuclear-dominated electricity sector keeps per capita emissions comparable to the UK, despite higher total energy consumption.

Key Takeaways

Emissions Are Still Rising — But the Map of Responsibility Is Shifting

01

Emissions Not Yet Peaked

Global annual CO_2 from fossil fuels and industry exceeds 35 billion tonnes. Growth has slowed but continues — no peak has been reached.

02

China Alone: >25% of Global Total

~12.5 Bt annually — more than double the US (~5 Bt). India (~3 Bt) is the third-largest and rising fast. Asia now generates ~50% of global emissions.